

Should-Cost & Make-vs-Buy Report

cube.step — 2026-10-01 · material class: polymer

Assumption-based should-cost. Every figure is a glass-box estimate built from stated drivers and rate assumptions (tagged DEFAULT / MODEL / SHOP / USER / MEASURED below). Confidence bands are **assumption-based, not yet validated** against measured shop outcomes — they are propagated from the stated assumptions, not a guarantee. Use as a negotiation and design baseline, not a quote.

Decision Governance

Status: unreviewed · **Recorded outcome:** Make outside · **Outcome basis:** mjf / PP (Polypropylene) / quantity 100 · **Outcome updated:** 2026-10-01T12:36:55.746168+00:00

Outcome note:
Audit 184 — MJF supplier at 100 parts; α/β <quoted> & reviewed. Note edited while viewing qty 1.

Geometry

Volume (cm³)	Surface area (cm²)	Bounding box (mm)	Watertight	Faces
2.72	14.32	20.0 × 15.0 × 10.0	Yes	185,328

Geometric Routing

Recommended process: **mjf** (archetype: bulk_solid, confidence 0.4)
General solid (91% of bbox filled, ~3.8mm mean wall) with no dominant sheet/rotational/prismatic signature → additive (MJF/SLS) for a polymer prototype.

Make-vs-Buy Crossover

Make now: fdm / PLA · **Tooling / production candidate:** injection_molding (redesign required)

Crossover quantity: ≈ 4592.0 units — above this the tooling/production route becomes the cheaper per-unit choice.

Lowest-cost no-tooling estimate at qty 1: fdm (PLA) — \$30.0/unit. Costed make-route picks: qty 1: fdm; qty 100: mjf; qty 1000: mjf; qty 2000: mjf; qty 10000: mjf; qty 12000: mjf. Estimated injection_molding crossover against the cheapest eligible no-tooling route: ~4592 units. Batch effects can shift this boundary; compare the costed options at your quantity. The tooling estimate is \$2.68/unit at qty 12000. Note: injection_molding requires redesign for DFM (114838 sidewall faces (100.0% of sidewall area) below 1.0° draft for injection_molding.); its cost is conditional on those changes, not a current-capability quote.

Recommended pick by quantity

Quantity	Process	Material	Unit cost	Cheaper if redesigned
1	fdm	PLA	\$30.00/unit	—
100	mjf	PP (Polypropylene)	\$3.62/unit	—
1000	mjf	PP (Polypropylene)	\$3.48/unit	—
2000	mjf	PP (Polypropylene)	\$3.48/unit	—
10000	mjf	PP (Polypropylene)	\$3.48/unit	injection_molding \$2.78/unit (add draft, tooling-dominated)
12000	mjf	PP (Polypropylene)	\$3.48/unit	injection_molding \$2.68/unit (add draft, tooling-dominated)

Per-Process Estimates (should-cost, USD)

Unit cost is the sum of the per-unit line items below, rounded to cents. One-time cost covers tooling and non-recurring work; long-run cost uses full-lot setup allocation. These two summary columns do not reproduce the exact unit cost at each quantity because batch rounding and minimum charges apply. The confidence band is **assumption-based, not yet validated** unless tagged **MEASURED**. "Validated?" is **No** for every assumption-band row.

Process	Material	Qty	Unit \$	One-time \$	Long-run \$/unit	Band (low-high)	±%	Validated?	DFM
cnc_3axis	Delrin (POM)	1	\$134.22	\$70.00	\$20.91	\$67.11–\$201.33	±50.0	No	Yes
cnc_3axis	Delrin (POM)	100	\$21.61	\$70.00	\$20.91	\$10.80–\$32.41	±50.0	No	Yes
cnc_3axis	Delrin (POM)	1000	\$14.94	\$70.00	\$14.87	\$7.47–\$22.41	±50.0	No	Yes
cnc_3axis	Delrin (POM)	2000	\$13.46	\$70.00	\$13.43	\$6.73–\$20.19	±50.0	No	Yes
cnc_3axis	Delrin (POM)	10000	\$10.62	\$70.00	\$10.61	\$5.31–\$15.93	±50.0	No	Yes
cnc_3axis	Delrin (POM)	12000	\$10.34	\$70.00	\$10.34	\$5.17–\$15.51	±50.0	No	Yes
cnc_5axis	Delrin (POM)	1	\$188.85	\$105.00	\$23.22	\$94.43–\$283.28	±50.0	No	Yes
cnc_5axis	Delrin (POM)	100	\$24.27	\$105.00	\$23.22	\$12.13–\$36.40	±50.0	No	Yes
cnc_5axis	Delrin (POM)	1000	\$16.65	\$105.00	\$16.55	\$8.33–\$24.98	±50.0	No	Yes
cnc_5axis	Delrin (POM)	2000	\$15.01	\$105.00	\$14.96	\$7.51–\$22.52	±50.0	No	Yes
cnc_5axis	Delrin (POM)	10000	\$11.86	\$105.00	\$11.85	\$5.93–\$17.79	±50.0	No	Yes
cnc_5axis	Delrin (POM)	12000	\$11.55	\$105.00	\$11.54	\$5.78–\$17.33	±50.0	No	Yes
d1p	Standard Resin	1	\$40.00	\$0.00	\$6.35	\$24.00–\$56.00	±40.0	No	Yes

Process	Material	Qty	Unit \$	One-time \$	Long-run \$/unit	Band (low-high)	±%	Validated?	DFM
dlp	Standard Resin	100	\$6.37	\$0.00	\$6.35	\$3.82–\$8.92	±40.0	No	Yes
dlp	Standard Resin	1000	\$6.35	\$0.00	\$6.35	\$3.81–\$8.89	±40.0	No	Yes
dlp	Standard Resin	2000	\$6.35	\$0.00	\$6.35	\$3.81–\$8.89	±40.0	No	Yes
dlp	Standard Resin	10000	\$6.35	\$0.00	\$6.35	\$3.81–\$8.89	±40.0	No	Yes
dlp	Standard Resin	12000	\$6.35	\$0.00	\$6.35	\$3.81–\$8.89	±40.0	No	Yes
fdm	PLA	1	\$30.00	\$0.00	\$8.54	\$18.00–\$42.00	±40.0	No	Yes
fdm	PLA	100	\$8.60	\$0.00	\$8.54	\$5.16–\$12.04	±40.0	No	Yes
fdm	PLA	1000	\$8.55	\$0.00	\$8.54	\$5.13–\$11.97	±40.0	No	Yes
fdm	PLA	2000	\$8.54	\$0.00	\$8.54	\$5.13–\$11.96	±40.0	No	Yes
fdm	PLA	10000	\$8.54	\$0.00	\$8.54	\$5.13–\$11.96	±40.0	No	Yes
fdm	PLA	12000	\$8.54	\$0.00	\$8.54	\$5.13–\$11.96	±40.0	No	Yes
injection_molding	PP (Molded)	1	\$6,002.18	\$6,000.00	\$2.18	\$2,400.87–\$9,603.48	±60.0	No	No
injection_molding	PP (Molded)	100	\$62.18	\$6,000.00	\$2.18	\$24.87–\$99.48	±60.0	No	No
injection_molding	PP (Molded)	1000	\$8.18	\$6,000.00	\$2.18	\$3.27–\$13.08	±60.0	No	No
injection_molding	PP (Molded)	2000	\$5.18	\$6,000.00	\$2.18	\$2.07–\$8.28	±60.0	No	No
injection_molding	PP (Molded)	10000	\$2.78	\$6,000.00	\$2.18	\$1.11–\$4.44	±60.0	No	No
injection_molding	PP (Molded)	12000	\$2.68	\$6,000.00	\$2.18	\$1.07–\$4.28	±60.0	No	No
mjf	PP (Polypropylene)	1	\$75.00	\$0.00	\$3.48	\$45.00–\$105.00	±40.0	No	Yes
mjf	PP (Polypropylene)	100	\$3.62	\$0.00	\$3.48	\$2.17–\$5.07	±40.0	No	Yes
mjf	PP (Polypropylene)	1000	\$3.48	\$0.00	\$3.48	\$2.09–\$4.88	±40.0	No	Yes
mjf	PP (Polypropylene)	2000	\$3.48	\$0.00	\$3.48	\$2.09–\$4.88	±40.0	No	Yes
mjf	PP (Polypropylene)	10000	\$3.48	\$0.00	\$3.48	\$2.09–\$4.88	±40.0	No	Yes
mjf	PP (Polypropylene)	12000	\$3.48	\$0.00	\$3.48	\$2.09–\$4.87	±40.0	No	Yes
sla	Standard Resin	1	\$40.00	\$0.00	\$10.44	\$24.00–\$56.00	±40.0	No	Yes
sla	Standard Resin	100	\$10.44	\$0.00	\$10.44	\$6.27–\$14.62	±40.0	No	Yes
sla	Standard Resin	1000	\$10.44	\$0.00	\$10.44	\$6.27–\$14.62	±40.0	No	Yes
sla	Standard Resin	2000	\$10.44	\$0.00	\$10.44	\$6.27–\$14.62	±40.0	No	Yes
sla	Standard Resin	10000	\$10.44	\$0.00	\$10.44	\$6.27–\$14.62	±40.0	No	Yes
sla	Standard Resin	12000	\$10.44	\$0.00	\$10.44	\$6.27–\$14.62	±40.0	No	Yes
sls	PA12 (Nylon 12)	1	\$75.00	\$0.00	\$3.67	\$45.00–\$105.00	±40.0	No	Yes
sls	PA12 (Nylon 12)	100	\$3.82	\$0.00	\$3.67	\$2.29–\$5.35	±40.0	No	Yes
sls	PA12 (Nylon 12)	1000	\$3.68	\$0.00	\$3.67	\$2.21–\$5.16	±40.0	No	Yes
sls	PA12 (Nylon 12)	2000	\$3.68	\$0.00	\$3.67	\$2.21–\$5.15	±40.0	No	Yes
sls	PA12 (Nylon 12)	10000	\$3.67	\$0.00	\$3.67	\$2.20–\$5.14	±40.0	No	Yes
sls	PA12 (Nylon 12)	12000	\$3.67	\$0.00	\$3.67	\$2.20–\$5.14	±40.0	No	Yes

Band basis (assumption-based rows): ±50% stated assumption band (cycle-time / tooling defaults) propagated around the point estimate — no ground truth yet

Line items (USD/unit — sum to unit cost)

Process	Material	Qty	nre	labor	machine	material	inspection	consumables	amortized_fixed	min_charge_floor	Σ unit
cnc_3axis	Delrin (POM)	1	\$70.00	\$17.50	\$1.80	\$0.02	\$18.55	\$0.09	\$26.25	—	\$134.22
cnc_3axis	Delrin (POM)	100	\$0.70	\$17.50	\$1.80	\$0.02	\$1.23	\$0.09	\$0.26	—	\$21.61
cnc_3axis	Delrin (POM)	1000	\$0.07	\$12.33	\$1.27	\$0.02	\$0.91	\$0.06	\$0.26	—	\$14.94
cnc_3axis	Delrin (POM)	2000	\$0.04	\$11.10	\$1.14	\$0.02	\$0.84	\$0.06	\$0.26	—	\$13.46
cnc_3axis	Delrin (POM)	10000	\$0.01	\$8.69	\$0.90	\$0.02	\$0.70	\$0.04	\$0.26	—	\$10.62
cnc_3axis	Delrin (POM)	12000	\$0.01	\$8.45	\$0.87	\$0.02	\$0.68	\$0.04	\$0.26	—	\$10.34
cnc_5axis	Delrin (POM)	1	\$105.00	\$17.50	\$3.17	\$0.02	\$28.00	\$0.16	\$35.00	—	\$188.85
cnc_5axis	Delrin (POM)	100	\$1.05	\$17.50	\$3.17	\$0.02	\$2.01	\$0.16	\$0.35	—	\$24.27
cnc_5axis	Delrin (POM)	1000	\$0.10	\$12.33	\$2.23	\$0.02	\$1.50	\$0.11	\$0.35	—	\$16.65
cnc_5axis	Delrin (POM)	2000	\$0.05	\$11.10	\$2.01	\$0.02	\$1.37	\$0.10	\$0.35	—	\$15.01
cnc_5axis	Delrin (POM)	10000	\$0.01	\$8.69	\$1.57	\$0.02	\$1.13	\$0.08	\$0.35	—	\$11.86
cnc_5axis	Delrin (POM)	12000	\$0.01	\$8.45	\$1.53	\$0.02	\$1.11	\$0.08	\$0.35	—	\$11.55
dlp	Standard Resin	1	—	\$5.31	\$0.78	\$0.18	—	—	\$10.50	\$23.23	\$40.00
dlp	Standard Resin	100	—	\$5.31	\$0.78	\$0.18	—	—	\$0.10	—	\$6.37
dlp	Standard Resin	1000	—	\$5.31	\$0.78	\$0.18	—	—	\$0.08	—	\$6.35
dlp	Standard Resin	2000	—	\$5.31	\$0.78	\$0.18	—	—	\$0.08	—	\$6.35

Process	Material	Qty	nre	labor	machine	material	inspection	consumables	amortized_fixed	min_charge_floor	Σ unit
dlp	Standard Resin	10000	—	\$5.31	\$0.78	\$0.18	—	—	\$0.08	—	\$6.35
dlp	Standard Resin	12000	—	\$5.31	\$0.78	\$0.18	—	—	\$0.08	—	\$6.35
fdm	PLA	1	—	\$7.08	\$1.16	\$0.09	—	—	\$8.75	\$12.91	\$30.00
fdm	PLA	100	—	\$7.08	\$1.16	\$0.09	—	—	\$0.26	—	\$8.60
fdm	PLA	1000	—	\$7.08	\$1.16	\$0.09	—	—	\$0.21	—	\$8.55
fdm	PLA	2000	—	\$7.08	\$1.16	\$0.09	—	—	\$0.21	—	\$8.54
fdm	PLA	10000	—	\$7.08	\$1.16	\$0.09	—	—	\$0.20	—	\$8.54
fdm	PLA	12000	—	\$7.08	\$1.16	\$0.09	—	—	\$0.20	—	\$8.54
injection_molding	PP (Molded)	1	—	\$1.75	\$0.42	\$0.01	—	—	\$6,000.00	—	\$6,002.18
injection_molding	PP (Molded)	100	—	\$1.75	\$0.42	\$0.01	—	—	\$60.00	—	\$62.18
injection_molding	PP (Molded)	1000	—	\$1.75	\$0.42	\$0.01	—	—	\$6.00	—	\$8.18
injection_molding	PP (Molded)	2000	—	\$1.75	\$0.42	\$0.01	—	—	\$3.00	—	\$5.18
injection_molding	PP (Molded)	10000	—	\$1.75	\$0.42	\$0.01	—	—	\$0.60	—	\$2.78
injection_molding	PP (Molded)	12000	—	\$1.75	\$0.42	\$0.01	—	—	\$0.50	—	\$2.68
mjf	PP (Polypropylene)	1	—	\$2.83	\$0.61	\$0.01	—	—	\$17.50	\$54.05	\$75.00
mjf	PP (Polypropylene)	100	—	\$2.83	\$0.61	\$0.01	—	—	\$0.17	—	\$3.62
mjf	PP (Polypropylene)	1000	—	\$2.83	\$0.61	\$0.01	—	—	\$0.04	—	\$3.48
mjf	PP (Polypropylene)	2000	—	\$2.83	\$0.61	\$0.01	—	—	\$0.04	—	\$3.48
mjf	PP (Polypropylene)	10000	—	\$2.83	\$0.61	\$0.01	—	—	\$0.03	—	\$3.48
mjf	PP (Polypropylene)	12000	—	\$2.83	\$0.61	\$0.01	—	—	\$0.03	—	\$3.48
sla	Standard Resin	1	—	\$5.53	\$4.32	\$0.18	—	—	\$10.50	\$19.48	\$40.00
sla	Standard Resin	100	—	\$5.53	\$4.32	\$0.18	—	—	\$0.42	—	\$10.44
sla	Standard Resin	1000	—	\$5.53	\$4.32	\$0.18	—	—	\$0.42	—	\$10.44
sla	Standard Resin	2000	—	\$5.53	\$4.32	\$0.18	—	—	\$0.42	—	\$10.44
sla	Standard Resin	10000	—	\$5.53	\$4.32	\$0.18	—	—	\$0.42	—	\$10.44
sla	Standard Resin	12000	—	\$5.53	\$4.32	\$0.18	—	—	\$0.42	—	\$10.44
sls	PA12 (Nylon 12)	1	—	\$2.82	\$0.65	\$0.18	—	—	\$17.50	\$53.85	\$75.00
sls	PA12 (Nylon 12)	100	—	\$2.82	\$0.65	\$0.18	—	—	\$0.17	—	\$3.82
sls	PA12 (Nylon 12)	1000	—	\$2.82	\$0.65	\$0.18	—	—	\$0.04	—	\$3.68
sls	PA12 (Nylon 12)	2000	—	\$2.82	\$0.65	\$0.18	—	—	\$0.03	—	\$3.68
sls	PA12 (Nylon 12)	10000	—	\$2.82	\$0.65	\$0.18	—	—	\$0.02	—	\$3.67
sls	PA12 (Nylon 12)	12000	—	\$2.82	\$0.65	\$0.18	—	—	\$0.02	—	\$3.67

Cost Drivers & Provenance

DEFAULT generic book value · **MODEL** learned/estimated · **SHOP** your calibrated profile · **USER** override · **MEASURED** real held-out residuals. Provenance holds across all quantities; values shown at qty 1 (reference).

cnc_3axis / Delrin (POM)

Driver	Value	Provenance	Basis / source
material_cost	0.0244 \$	DEFAULT	bounding-box billet 3.00 cm³ (oriented candidate 10×15×20 mm; minimum not certified) × 1.10 stock allowance × Delrin (POM) density 1.41 g/cm³ [assumption, not shop-validated] = 0.0047 kg × \$5/kg (material-DB unit price (DEFAULT book value)) × (1+0.05 scrap) × region-material ×1
machine_cost	1.8045 \$	DEFAULT	0.0241 hr × \$75/hr × region-labor ×1 [rough 0.6 cm³ ÷ (50 cm³/min-60) = 0.000 hr + finish 14.3 cm² ÷ 600 cm²/hr = 0.024 hr [stock oriented candidate bbox billet 3.0 cm³ × 1.10 = 3.3 cm³ [assumption, not shop-validated]; MRR polymer]]
cycle_time	0.0241 hr	DEFAULT	rough 0.6 cm³ ÷ (50 cm³/min-60) = 0.000 hr + finish 14.3 cm² ÷ 600 cm²/hr = 0.024 hr [stock oriented candidate bbox billet 3.0 cm³ × 1.10 = 3.3 cm³ [assumption, not shop-validated]; MRR polymer]
labor_cost	17.5 \$	DEFAULT	post-process 0.5 hr × \$35/hr × region-labor ×1
setup_cost	26.25 \$	DEFAULT	setup 0.75hr × \$35/hr × ceil(1/100) = 1 setups ÷ 1 × region-labor ×1
nre_cost	70.0 \$	DEFAULT	CAM programming/NRE 2hr × \$35/hr × region-labor ×1 ÷ 1 order (one-time) [assumption, not shop-validated]
inspection_cost	18.55 \$	DEFAULT	first-article 0.5hr × 1 lot(s) ÷ 1 + in-process 0.03hr/part = 0.5300hr × \$35/hr × region-labor ×1 [assumption, not shop-validated]
consumables_cost	0.0902 \$	DEFAULT	perishable tooling/consumables 0.05 × machine \$1.80 (cutting tools, inserts, coolant) [assumption, not shop-validated]

cnc_5axis / Delrin (POM)

Driver	Value	Provenance	Basis / source
material_cost	0.0244 \$	DEFAULT	bounding-box billet 3.00 cm³ (oriented candidate 10×15×20 mm; minimum not certified) × 1.10 stock allowance × Delrin (POM) density 1.41 g/cm³ [assumption, not shop-validated] = 0.0047 kg × \$5/kg (material-DB unit price (DEFAULT book value)) × (1+0.05 scrap) × region-material ×1
machine_cost	3.1717 \$	DEFAULT	0.0288 hr × \$110/hr × region-labor ×1 [rough 0.6 cm³ ÷ (50 cm³/min-60) = 0.000 hr + finish 14.3 cm² ÷ 500 cm²/hr = 0.029 hr [stock oriented candidate bbox billet 3.0 cm³ × 1.10 = 3.3 cm³ [assumption, not shop-validated]; MRR polymer]]
cycle_time	0.0288 hr	DEFAULT	rough 0.6 cm³ ÷ (50 cm³/min-60) = 0.000 hr + finish 14.3 cm² ÷ 500 cm²/hr = 0.029 hr [stock oriented candidate bbox billet 3.0 cm³ × 1.10 = 3.3 cm³ [assumption, not shop-validated]; MRR polymer]
labor_cost	17.5 \$	DEFAULT	post-process 0.5 hr × \$35/hr × region-labor ×1
setup_cost	35.0 \$	DEFAULT	setup 1hr × \$35/hr × ceil(1/100) = 1 setups ÷ 1 × region-labor ×1
nre_cost	105.0 \$	DEFAULT	CAM programming/NRE 3hr × \$35/hr × region-labor ×1 ÷ 1 order (one-time) [assumption, not shop-validated]
inspection_cost	28.0 \$	DEFAULT	first-article 0.75hr × 1 lot(s) ÷ 1 + in-process 0.05hr/part = 0.8000hr × \$35/hr × region-labor ×1 [assumption, not shop-validated]
consumables_cost	0.1586 \$	DEFAULT	perishable tooling/consumables 0.05 × machine \$3.17 (cutting tools, inserts, coolant) [assumption, not shop-validated]

dlp / Standard Resin

Driver	Value	Provenance	Basis / source
material_cost	0.1764 \$	DEFAULT	CAD volume 2.72 cm³ × Standard Resin density 1.18 g/cm³ = 0.0032 kg × \$50/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1
parts_per_build	125.0 parts	DEFAULT	build-job 245mm ÷ 30mm/hr = 8.2hr full build ÷ 125 parts/build (packing 0.12, env (192, 120, 245)) = 0.065hr/part; oriented candidate oriented XYZ (15.0, 20.0, 10.0)mm; spaced-grid capacity limit (packing/setup assumption, not a verified nesting plan)
machine_cost	0.784 \$	DEFAULT	0.0653 hr × \$12/hr × region-labor ×1 [build-job 245mm ÷ 30mm/hr = 8.2hr full build ÷ 125 parts/build (packing 0.12, env (192, 120, 245)) = 0.065hr/part; oriented candidate oriented XYZ (15.0, 20.0, 10.0)mm; spaced-grid capacity limit (packing/setup assumption, not a verified nesting plan)]
cycle_time	0.0653 hr	DEFAULT	build-job 245mm ÷ 30mm/hr = 8.2hr full build ÷ 125 parts/build (packing 0.12, env (192, 120, 245)) = 0.065hr/part; oriented candidate oriented XYZ (15.0, 20.0, 10.0)mm; spaced-grid capacity limit (packing/setup assumption, not a verified nesting plan)
labor_cost	5.306 \$	DEFAULT	finish 0.15hr/part + bulk 0.2hr/build ÷ 125 = 0.1516hr × \$35/hr × region-labor ×1
setup_cost	10.5 \$	DEFAULT	setup 0.3hr × \$35/hr × ceil(1/125) = 1 setups ÷ 1 × region-labor ×1
min_charge_floor	23.2336 \$	DEFAULT	shop/order minimum \$40/lot × 1 lots ÷ 1 = \$40.00/unit floor (applied)

fdm / PLA

Driver	Value	Provenance	Basis / source
material_cost	0.0927 \$	DEFAULT	CAD volume 2.72 cm³ × PLA density 1.24 g/cm³ = 0.0034 kg × \$25/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1
parts_per_build	43.0 parts	DEFAULT	serial XY nest: deposition V/16 = 2.72/16 = 0.170hr/part (per-part nozzle) + Z-sweep (10.0/25)÷43 parts/plate = 0.009hr/part (plate Z-climb amortized; XY packing 0.5, plate 200×200mm) = 0.179hr/part; oriented candidate oriented XYZ (15.0, 20.0, 10.0)mm; spaced-grid capacity limit (packing/setup assumption, not a verified nesting plan); USER-declared envelope of 'Audit local FDM 046'
machine_cost	1.1644 \$	USER	0.1791 hr × \$10/hr (fitted machine 'Audit local FDM 046', its OWN declared rate) × (1-0.35 capital sunk) OWNED-IN-HOUSE marginal [assumption, not shop-validated] × region-labor ×1 [serial XY nest: deposition V/16 = 2.72/16 = 0.170hr/part (per-part nozzle) + Z-sweep (10.0/25)÷43 parts/plate = 0.009hr/part (plate Z-climb amortized; XY packing 0.5, plate 200×200mm) = 0.179hr/part; oriented candidate oriented XYZ (15.0, 20.0, 10.0)mm; spaced-grid capacity limit (packing/setup assumption, not a verified nesting plan); USER-declared envelope of 'Audit local FDM 046']
owned_in_house	0.65 ×	USER	machine 'Audit local FDM 046' OWNED in-house: machine costed at MARGINAL rate \$6.5/hr = \$10/hr × (1-0.35 capital) — capital purchase/amortization removed (sunk on gear the org already owns; make-it-ourselves, not rent-the-time). USER-declared ownership; the 0.35 capital fraction is DEFAULT [assumption, not shop-validated]
cycle_time	0.1791 hr	DEFAULT	serial XY nest: deposition V/16 = 2.72/16 = 0.170hr/part (per-part nozzle) + Z-sweep (10.0/25)÷43 parts/plate = 0.009hr/part (plate Z-climb amortized; XY packing 0.5, plate 200×200mm) = 0.179hr/part; oriented candidate oriented XYZ (15.0, 20.0, 10.0)mm; spaced-grid capacity limit (packing/setup assumption, not a verified nesting plan); USER-declared envelope of 'Audit local FDM 046'
labor_cost	7.0814 \$	DEFAULT	finish 0.2hr/part + bulk 0.1hr/build ÷ 43 ≈ 0.202326hr × \$35/hr × region-labor ×1
setup_cost	8.75 \$	DEFAULT	setup 0.25hr × \$35/hr × ceil(1/43) = 1 setups ÷ 1 × region-labor ×1
min_charge_floor	12.9115 \$	DEFAULT	shop/order minimum \$30/lot × 1 lots ÷ 1 = \$30.00/unit floor (applied)

injection_molding / PP (Molded)

Driver	Value	Provenance	Basis / source
material_cost	0.005 \$	DEFAULT	CAD volume 2.72 cm³ × PP (Molded) density 0.90 g/cm³ = 0.0024 kg × \$2/kg (material-DB unit price (DEFAULT book value)) × (1+0.03 scrap) × region-material ×1
machine_cost	0.4226 \$	DEFAULT	0.0094 hr × \$45/hr × region-labor ×1 [cooling 2-wall² = 2·3.80² = 28.8s + shot 5s = 33.8s = 0.0094 hr (wall = 2V/A proxy, ±50%)]
cycle_time	0.0094 hr	DEFAULT	cooling 2-wall² = 2·3.80² = 28.8s + shot 5s = 33.8s = 0.0094 hr (wall = 2V/A proxy, ±50%)
labor_cost	1.75 \$	DEFAULT	post-process 0.05 hr × \$35/hr × region-labor ×1
setup_cost	0.0 \$	DEFAULT	setup 0hr × \$35/hr × ceil(1/100000) = 1 setups ÷ 1 × region-labor ×1
tooling_cost	6000.0 \$	DEFAULT	size tier S (max bbox 20mm) × 1 cav^0.7 (=1.00) × moderate (=1.00) = \$6,000; ±60%, OVERRIDABLE

mjf / PP (Polypropylene)

Driver	Value	Provenance	Basis / source
material_cost	0.0054 \$	DEFAULT	CAD volume 2.72 cm³ × PP (Polypropylene) density 0.90 g/cm³ = 0.0024 kg × \$2/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1
parts_per_build	546.0 parts	DEFAULT	build-job 380mm ÷ 25mm/hr = 15.2hr full build ÷ 546 parts/build (packing 0.1, env (380, 284, 380)) = 0.028hr/part; oriented candidate oriented XYZ (15.0, 20.0, 10.0)mm; spaced-grid capacity limit (packing/setup assumption, not a verified nesting plan)
machine_cost	0.6125 \$	DEFAULT	0.0278 hr × \$22/hr × region-labor ×1 [build-job 380mm ÷ 25mm/hr = 15.2hr full build ÷ 546 parts/build (packing 0.1, env (380, 284, 380)) = 0.028hr/part; oriented candidate oriented XYZ (15.0, 20.0, 10.0)mm; spaced-grid capacity limit (packing/setup assumption, not a verified nesting plan)]
cycle_time	0.0278 hr	DEFAULT	build-job 380mm ÷ 25mm/hr = 15.2hr full build ÷ 546 parts/build (packing 0.1, env (380, 284, 380)) = 0.028hr/part; oriented candidate oriented XYZ (15.0, 20.0, 10.0)mm; spaced-grid capacity limit (packing/setup assumption, not a verified nesting plan)
labor_cost	2.8321 \$	DEFAULT	finish 0.08hr/part + bulk 0.5hr/build ÷ 546 ≈ 0.0809158hr × \$35/hr × region-labor ×1
setup_cost	17.5 \$	DEFAULT	setup 0.5hr × \$35/hr × ceil(1/546) = 1 setups ÷ 1 × region-labor ×1
min_charge_floor	54.05 \$	DEFAULT	shop/order minimum \$75/lot × 1 lots ÷ 1 = \$75.00/unit floor (applied)

sla / Standard Resin

Driver	Value	Provenance	Basis / source
material_cost	0.1764 \$	DEFAULT	CAD volume 2.72 cm³ × Standard Resin density 1.18 g/cm³ = 0.0032 kg × \$50/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1
parts_per_build	25.0 parts	DEFAULT	serial XY nest: deposition V/8 = 2.72/8 = 0.340hr/part (per-part nozzle) + Z-sweep (10.0/20)÷25 parts/plate = 0.020hr/part (plate Z-climb amortized; XY packing 0.5, plate 145×145mm) = 0.360hr/part; oriented candidate oriented XYZ (15.0, 20.0, 10.0)mm; spaced-grid capacity limit (packing/setup assumption, not a verified nesting plan)
machine_cost	4.316 \$	DEFAULT	0.3597 hr × \$12/hr × region-labor ×1 [serial XY nest: deposition V/8 = 2.72/8 = 0.340hr/part (per-part nozzle) + Z-sweep (10.0/20)÷25 parts/plate = 0.020hr/part (plate Z-climb amortized; XY packing 0.5, plate 145×145mm) = 0.360hr/part; oriented candidate oriented XYZ (15.0, 20.0, 10.0)mm; spaced-grid capacity limit (packing/setup assumption, not a verified nesting plan)]
cycle_time	0.3597 hr	DEFAULT	serial XY nest: deposition V/8 = 2.72/8 = 0.340hr/part (per-part nozzle) + Z-sweep (10.0/20)÷25 parts/plate = 0.020hr/part (plate Z-climb amortized; XY packing 0.5, plate 145×145mm) = 0.360hr/part; oriented candidate oriented XYZ (15.0, 20.0, 10.0)mm; spaced-grid capacity limit (packing/setup assumption, not a verified nesting plan)
labor_cost	5.53 \$	DEFAULT	finish 0.15hr/part + bulk 0.2hr/build ÷ 25 ≈ 0.158hr × \$35/hr × region-labor ×1
setup_cost	10.5 \$	DEFAULT	setup 0.3hr × \$35/hr × ceil(1/25) = 1 setups ÷ 1 × region-labor ×1
min_charge_floor	19.4776 \$	DEFAULT	shop/order minimum \$40/lot × 1 lots ÷ 1 = \$40.00/unit floor (applied)

sls / PA12 (Nylon 12)

Driver	Value	Provenance	Basis / source
material_cost	0.1811 \$	DEFAULT	CAD volume 2.72 cm³ × PA12 (Nylon 12) density 1.01 g/cm³ = 0.0027 kg × \$60/kg (material-DB unit price (DEFAULT book value)) × (1+0.1 scrap) × region-material ×1
parts_per_build	924.0 parts	DEFAULT	build-job 600mm ÷ 20mm/hr = 30.0hr full build ÷ 924 parts/build (packing 0.1, env (340, 340, 600)) = 0.032hr/part; oriented candidate oriented XYZ (15.0, 20.0, 10.0)mm; spaced-grid capacity limit (packing/setup assumption, not a verified nesting plan)
machine_cost	0.6494 \$	DEFAULT	0.0325 hr × \$20/hr × region-labor ×1 [build-job 600mm ÷ 20mm/hr = 30.0hr full build ÷ 924 parts/build (packing 0.1, env (340, 340, 600)) = 0.032hr/part; oriented candidate oriented XYZ (15.0, 20.0, 10.0)mm; spaced-grid capacity limit (packing/setup assumption, not a verified nesting plan)]
cycle_time	0.0325 hr	DEFAULT	build-job 600mm ÷ 20mm/hr = 30.0hr full build ÷ 924 parts/build (packing 0.1, env (340, 340, 600)) = 0.032hr/part; oriented candidate oriented XYZ (15.0, 20.0, 10.0)mm; spaced-grid capacity limit (packing/setup assumption, not a verified nesting plan)
labor_cost	2.8189 \$	DEFAULT	finish 0.08hr/part + bulk 0.5hr/build ÷ 924 ≈ 0.0805411hr × \$35/hr × region-labor ×1
setup_cost	17.5 \$	DEFAULT	setup 0.5hr × \$35/hr × ceil(1/924) = 1 setups ÷ 1 × region-labor ×1
min_charge_floor	53.8506 \$	DEFAULT	shop/order minimum \$75/lot × 1 lots ÷ 1 = \$75.00/unit floor (applied)

Assumptions Log

DEFAULT = generic fallback · SHOP = your calibrated profile · USER = ad-hoc override. Every assumption is editable and re-costable.

Assumption	Value	Unit	Source	Provenance
labor_rate	35.0	\$/hr	loaded shop-floor labor	DEFAULT
region_labor	1.0	x	region US: labor/machine/setup ×1	DEFAULT
region_material	1.0	x	region US: commodity material ×1 (global feedstock, ~1.0)	DEFAULT
region_tooling	1.0	x	region US: tooling ×1 (offshore toolmaking labor)	DEFAULT
margin	0.0	frac	markup on cost: price = cost × (1 + margin); 0.25 adds 25%, not a 25% gross margin	DEFAULT
overhead	0.0	frac	indirect burden on conversion cost (machine+labor+setup)	DEFAULT
utilization	1.0	frac	machine utilization (idle-recovery on machine cost)	DEFAULT
stock_allowance	1.1	x	CNC stock volume allowance (milling: on bounding box; turning: on enclosing cylinder)	DEFAULT
machine_labor_frac	0.35	frac	operator-labor share of the machine rate that scales with region (capital/facility stays global) [assumption, not shop-validated]	DEFAULT
perishable_frac	0.05	frac	perishable tooling/consumables as a fraction of CNC machine cost [assumption, not shop-validated]	DEFAULT
daily_machine_hours	8.0	hr/day	fallback daily capacity; process-specific machine hours take precedence	DEFAULT
n_cavities	1.0	cav	formative tooling cavities = 1 (DEFAULT single-cavity should-cost)	DEFAULT
complexity	0.0	moderate	tooling complexity = moderate (DEFAULT moderate; raise for slides/side-actions)	DEFAULT
material_class	0.0	polymer	material class = polymer (DEFAULT: these are polymer automotive parts; override for metal)	DEFAULT
machine_capital_frac	0.35	frac	capital/amortization share of the loaded machine rate removed for OWNED, in-house processes (fdm) — sunk on gear the org already owns [assumption, not shop-validated]	DEFAULT

Notes

- DFM note: processes injection_molding are costed but flagged NOT DFM-ready as-modeled (see their process-specific blockers). Their cost shows the tooling economics "if" the part is redesigned for that process. Set strict_dfm=True to exclude them entirely.
- Geometric routing: this part reads as a 'bulk_solid' → mjf. General solid (91% of bbox filled, ~3.8mm mean wall) with no dominant sheet/rotational/prismatic signature → additive (MJF/SLS) for a polymer prototype.
- Note: the cost-cheapest make-as-is pick at low qty is fdm, but the geometry-recommended process is mjf — both are costed above; pick on intent, not just the marginal \$.
- Prices and crossover quantities are assumption-based and can shift with cycle times, tooling, batch sizes and shop rates. Validate against actual quotes before making a sourcing commitment.